

Import Data

```
In [1]: import pandas as pd  
import seaborn as sns  
import matplotlib.pyplot as plt  
%matplotlib inline
```

```
In [9]: df = pd.read_csv(r"D:\Downloads\Udemy-PY\HR_Data_MNC_Data Science Lovers.csv")
```

```
In [10]: df
```

Out[10]:

Unnamed: 0	Employee_ID	Full_Name	Department	Job_Title	Hire_Date	
0	0	EMP0000001	Joshua Nguyen	IT	Software Engineer	2011-08-10
1	1	EMP0000002	Julie Williams	Marketing	SEO Specialist	2018-03-02
2	2	EMP0000003	Alyssa Martinez	HR	HR Manager	2023-03-20
3	3	EMP0000004	Nicholas Valdez	IT	Software Engineer	2023-10-12
4	4	EMP0000005	Joel Hendricks	Operations	Logistics Coordinator	2024-12-09
...	...	...	...	...	...	...
1999995	1999995	EMP1999996	Cody Russell	Operations	Logistics Coordinator	2010-08-31
1999996	1999996	EMP1999997	Tracey Smith	IT	Software Engineer	2021-05-07
1999997	1999997	EMP1999998	Tracy Lee	Sales	Business Development Manager	2024-05-29
1999998	1999998	EMP1999999	Michael Roberson	IT	Software Engineer	2023-02-14
1999999	1999999	EMP2000000	Angela Lambert	HR	Talent Acquisition Specialist	2020-11-11

2000000 rows × 12 columns



In [11]:

```
df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2000000 entries, 0 to 1999999
Data columns (total 12 columns):
 #   Column                Dtype
---  -
 0   Unnamed: 0            int64
 1   Employee_ID           object
 2   Full_Name            object
 3   Department           object
 4   Job_Title            object
 5   Hire_Date            object
 6   Location             object
 7   Performance_Rating   int64
 8   Experience_Years     int64
 9   Status              object
10   Work_Mode            object
11   Salary_INR          int64
dtypes: int64(4), object(8)
memory usage: 183.1+ MB

```

```
In [12]: df.drop(columns = 'Unnamed: 0', inplace = True)           #Drop unwanted column
```

```
In [13]: df.head(2)
```

```
Out[13]:
```

	Employee_ID	Full_Name	Department	Job_Title	Hire_Date	Location	Performance_F
--	-------------	-----------	------------	-----------	-----------	----------	---------------

0	EMP0000001	Joshua Nguyen	IT	Software Engineer	2011-08- 10	Isaacland, Denmark	
---	------------	------------------	----	----------------------	----------------	-----------------------	--

1	EMP0000002	Julie Williams	Marketing	SEO Specialist	2018-03- 02	Anthony'side, Costa Rica	
---	------------	-------------------	-----------	-------------------	----------------	-----------------------------	--



```
In [14]: df['Hire_Date'] = pd.to_datetime(df['Hire_Date'])
```

```
In [15]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2000000 entries, 0 to 1999999
Data columns (total 11 columns):
 #   Column                Dtype
---  -
 0   Employee_ID           object
 1   Full_Name             object
 2   Department            object
 3   Job_Title             object
 4   Hire_Date             datetime64[ns]
 5   Location              object
 6   Performance_Rating    int64
 7   Experience_Years      int64
 8   Status                object
 9   Work_Mode             object
10   Salary_INR            int64
dtypes: datetime64[ns](1), int64(3), object(7)
memory usage: 167.8+ MB
```

```
In [30]: sns.countplot(x = 'Experience_Years', data = df, palette = ['yellow','blue','navy'])
plt.show();
```

C:\Users\hmmuz\AppData\Local\Temp\ipykernel_16380\877648922.py:1: FutureWarning:

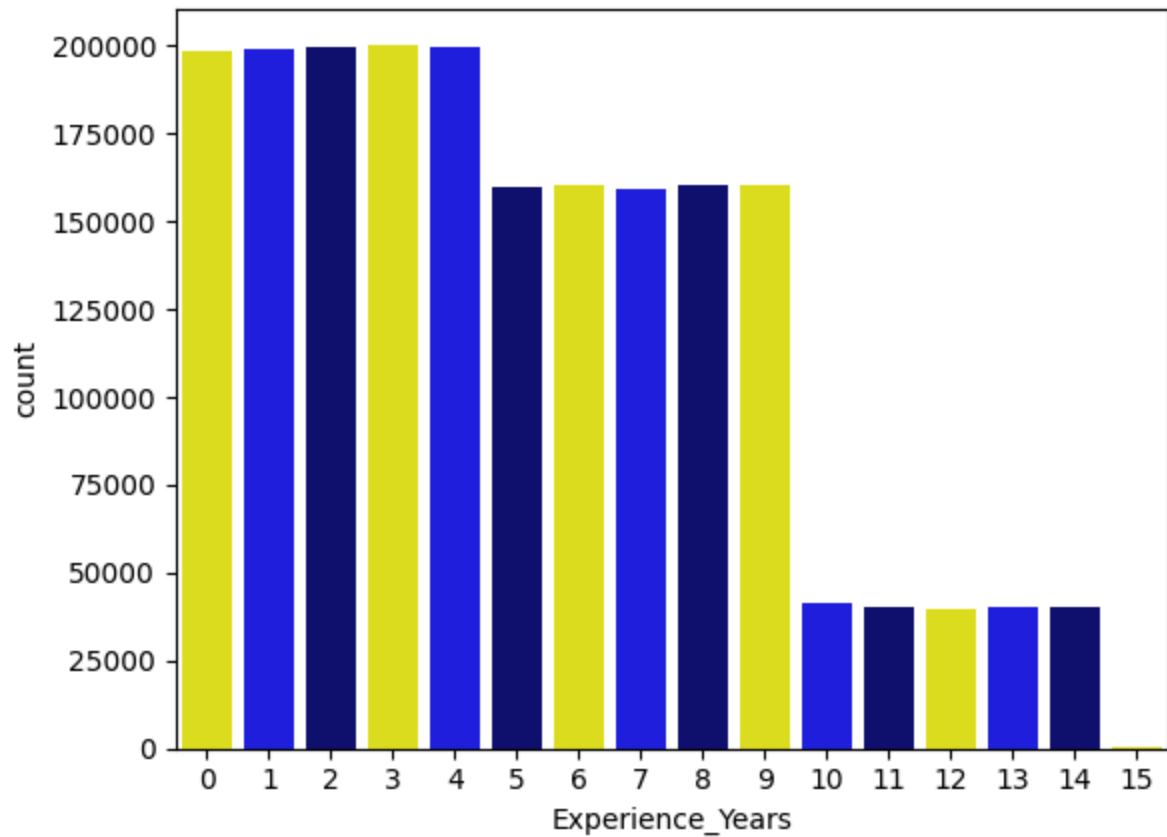
Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.countplot(x = 'Experience_Years', data = df, palette = ['yellow','blue','nav
y']) # Experienced Employees
```

C:\Users\hmmuz\AppData\Local\Temp\ipykernel_16380\877648922.py:1: UserWarning:

The palette list has fewer values (3) than needed (16) and will cycle, which may produce an uninterpretable plot.

```
sns.countplot(x = 'Experience_Years', data = df, palette = ['yellow','blue','nav
y']) # Experienced Employees
```

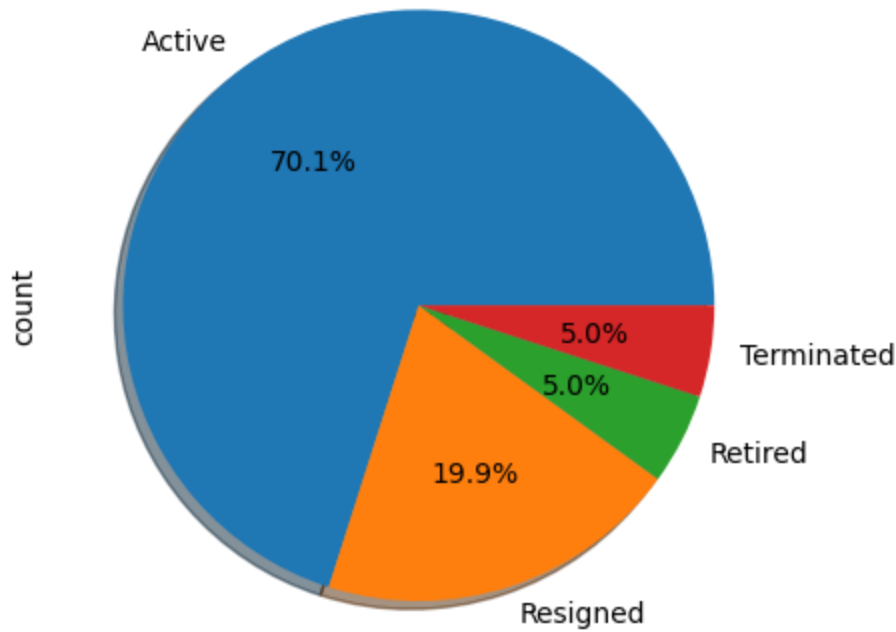


1. What is the distribution of Employees? (Resigned, Active, etc.)

```
In [33]: status = df['Status'].value_counts()  
status
```

```
Out[33]: Status  
Active      1401558  
Resigned    398660  
Retired      99912  
Terminated   99870  
Name: count, dtype: int64
```

```
In [37]: status.plot(kind = 'pie', shadow = True, autopct = '%1.1f%');
```

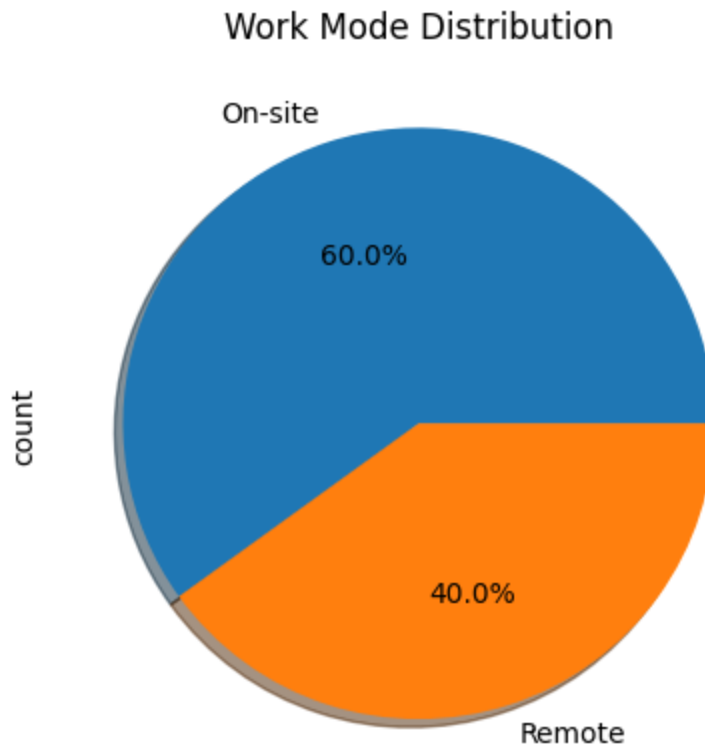


2. What is the distribution of workmodes? (Onsite, Remote etc.)

```
In [39]: work = df['Work_Mode'].value_counts()  
work
```

```
Out[39]: Work_Mode  
On-site    1199109  
Remote      800891  
Name: count, dtype: int64
```

```
In [42]: work.plot(kind = 'pie', shadow = True, autopct = '%1.1f%')  
plt.title('Work Mode Distribution')  
plt.show();
```



3. How much Employees are there in each Department?

```
In [44]: dept = df['Department'].value_counts()  
dept
```

```
Out[44]: Department  
IT          601042  
Sales       400031  
Operations  300095  
Marketing   240081  
Finance     199873  
HR          159119  
R&D         99759  
Name: count, dtype: int64
```

```
In [52]: sns.countplot(x= 'Department', data = df, palette = ['navy','yellow', 'lightgreen']  
plt.xticks(rotation = 'vertical')  
plt.show()
```

C:\Users\hmmuz\AppData\Local\Temp\ipykernel_16380\3289437392.py:1: FutureWarning:

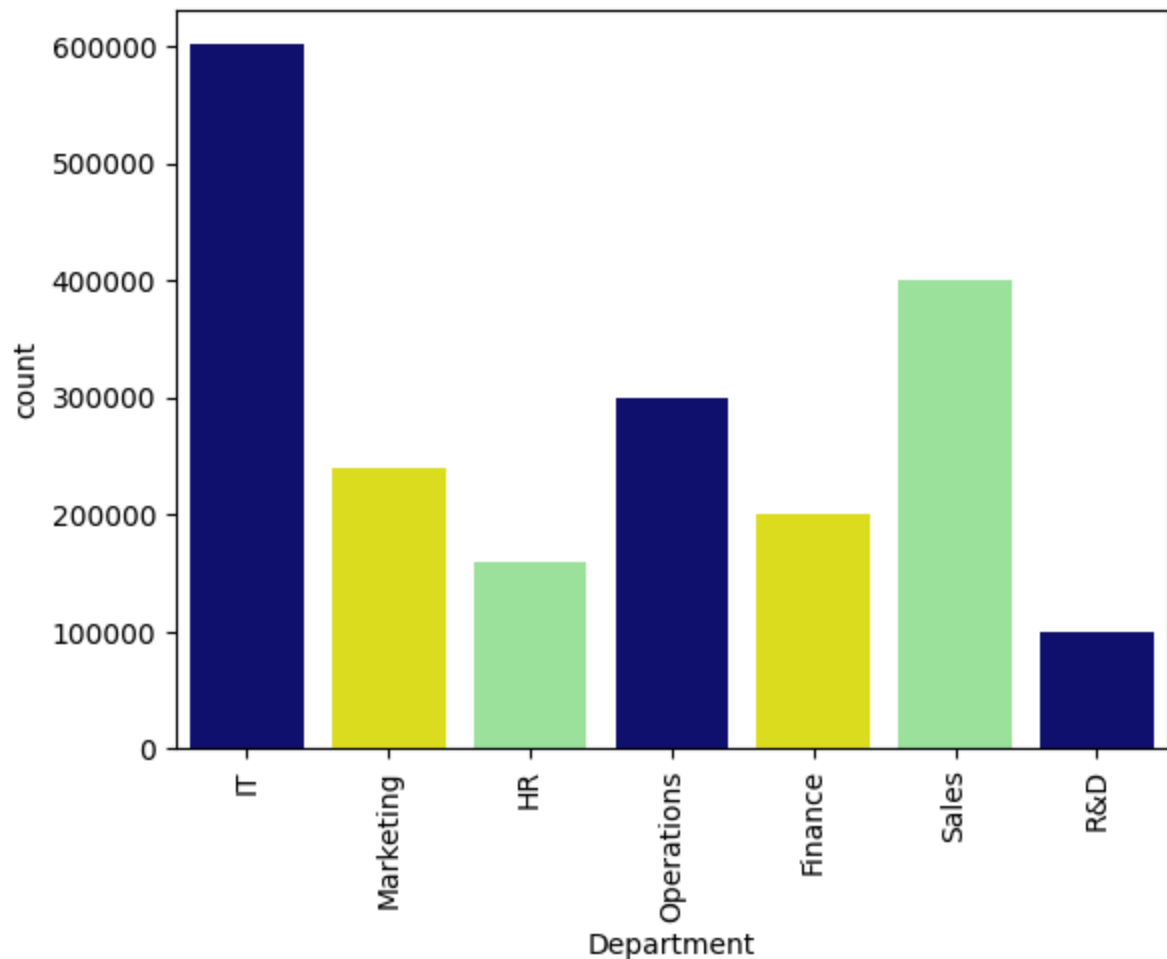
Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.countplot(x= 'Department', data = df, palette = ['navy','yellow', 'lightgreen'])
```

C:\Users\hmmuz\AppData\Local\Temp\ipykernel_16380\3289437392.py:1: UserWarning:

The palette list has fewer values (3) than needed (7) and will cycle, which may produce an uninterpretable plot.

```
sns.countplot(x= 'Department', data = df, palette = ['navy','yellow', 'lightgreen'])
```



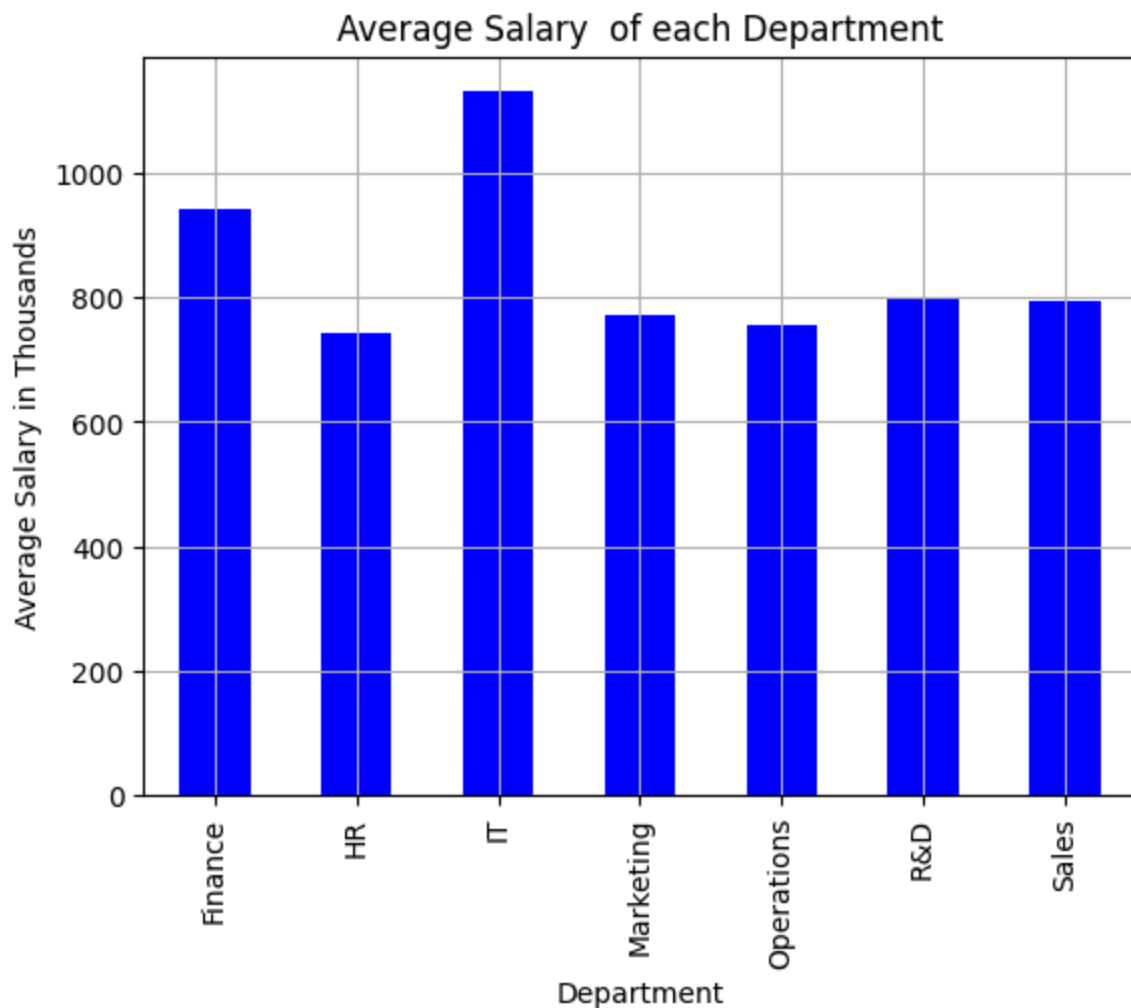
4. What is the average Salary of each Department?

```
In [63]: salary = df.groupby('Department')['Salary_INR'].mean()/1000
salary
```



```
Out[63]: Department
Finance      940.411743
HR           743.853561
IT           1129.858151
Marketing     769.936152
Operations    754.626253
R&D          800.377157
Sales        792.957860
Name: Salary_INR, dtype: float64
```

```
In [67]: salary.plot( x = salary.index, y = salary.values, kind = 'bar', color = 'blue')
plt.title('Average Salary of each Department')
plt.xlabel('Department')
plt.ylabel('Average Salary in Thousands')
plt.grid()
plt.show();
```



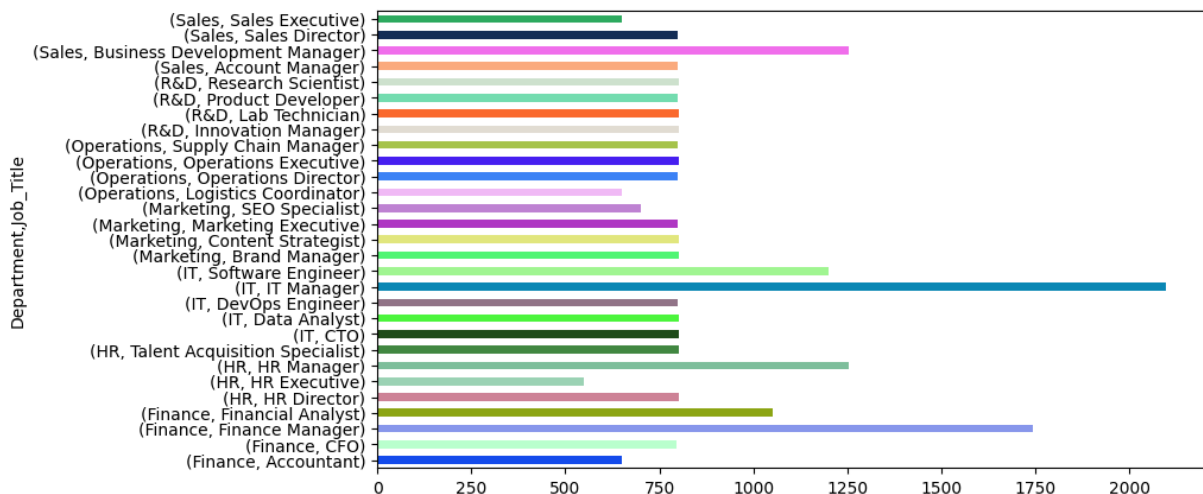
5. What is the Average Salary of each Department based on the Job title?

```
In [70]: avg = df.groupby(['Department', 'Job_Title'])['Salary_INR'].mean()/1000
avg
```

```
Out[70]: Department Job_Title
Finance Accountant 650.076482
          CFO 795.015873
          Finance Manager 1743.241525
          Financial Analyst 1051.522903
HR HR Director 800.694437
   HR Executive 550.548859
   HR Manager 1252.401915
   Talent Acquisition Specialist 801.422237
IT CTO 801.402754
   Data Analyst 800.996380
   DevOps Engineer 799.949184
   IT Manager 2098.155777
   Software Engineer 1199.260843
Marketing Brand Manager 803.127787
          Content Strategist 800.760030
          Marketing Executive 798.780404
          SEO Specialist 700.456337
Operations Logistics Coordinator 649.631726
          Operations Director 798.298093
          Operations Executive 800.350915
          Supply Chain Manager 798.168555
R&D Innovation Manager 801.870103
    Lab Technician 800.181468
    Product Developer 798.652261
    Research Scientist 801.314879
Sales Account Manager 799.373734
      Business Development Manager 1252.016231
      Sales Director 799.069374
      Sales Executive 650.237755
Name: Salary_INR, dtype: float64
```

```
In [95]: import random
numBars = len(avg)
random_colors = [f'#{random.randint(0, 0xFFFFFF):06x}' for _ in range(numBars)]

avg.plot(kind = 'barh', figsize = (9,5), color = random_colors )
plt.show();
```



6. How Salary varies on Experience?

In [97]: `df.head(2)`

Out[97]:

	Employee_ID	Full_Name	Department	Job_Title	Hire_Date	Location	Performance_F
0	EMP0000001	Joshua Nguyen	IT	Software Engineer	2011-08- 10	Isaacland, Denmark	
1	EMP0000002	Julie Williams	Marketing	SEO Specialist	2018-03- 02	Anthony'side, Costa Rica	

In [99]: `df.groupby('Experience_Years')['Salary_INR'].mean()`

Out[99]:

Experience_Years	
0	896737.454775
1	895903.759824
2	896755.652313
3	896861.245240
4	897944.573965
5	896484.084828
6	896012.632467
7	895722.673960
8	897148.361090
9	898482.940577
10	895662.027882
11	901452.750112
12	896432.933416
13	898790.197041
14	895610.790251
15	895647.401051

Name: Salary_INR, dtype: float64

7. What is the Average performance Rating by Department?

In [101...]: `pr = df.groupby('Department')['Performance_Rating'].mean()
pr`

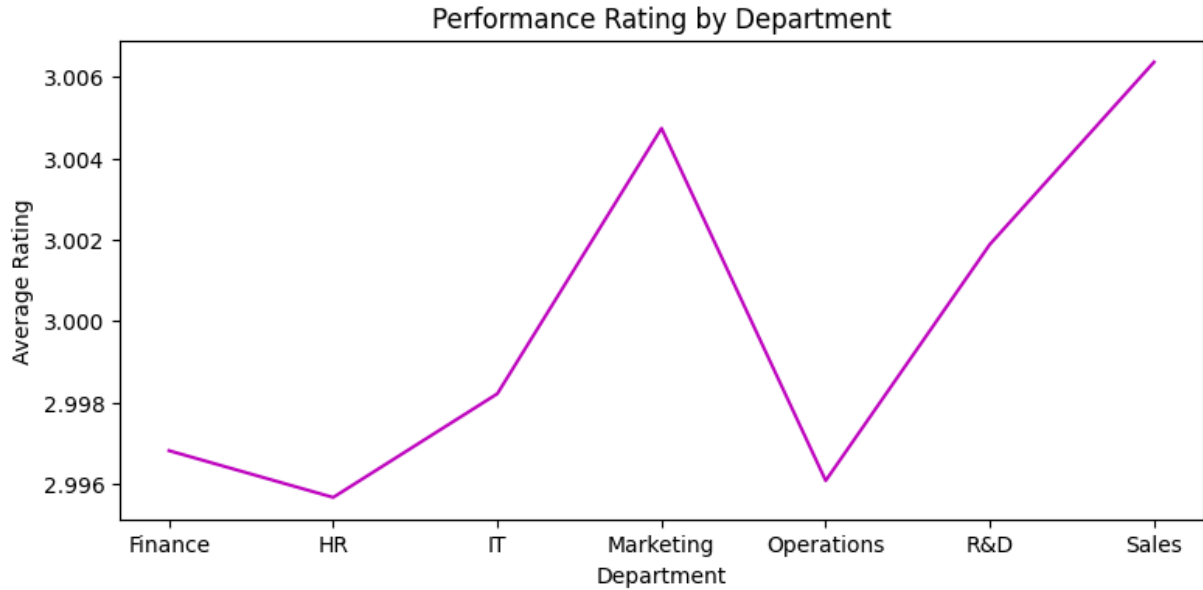
Out[101...]:

Department	
Finance	2.996818
HR	2.995670
IT	2.998216
Marketing	3.004736
Operations	2.996081
R&D	3.001885
Sales	3.006362

Name: Performance_Rating, dtype: float64

In [105...]: `pr.plot(x = 'Performance_Rating', figsize = (9,4), color = 'm')
plt.title('Performance Rating by Department')`

```
plt.ylabel('Average Rating')
plt.show();
```



8. Which Country has highest concentration of Employees?

In [106... `df.head(2)`

Out[106...]

	Employee_ID	Full_Name	Department	Job_Title	Hire_Date	Location	Performance_F
0	EMP0000001	Joshua Nguyen	IT	Software Engineer	2011-08-10	Isaacland, Denmark	
1	EMP0000002	Julie Williams	Marketing	SEO Specialist	2018-03-02	Anthony'side, Costa Rica	

In [107... `df['Country'] = df['Location'].apply(lambda x : str(x.split(',')[1]))`

In [108... `df.head()`

Out[108...

	Employee_ID	Full_Name	Department	Job_Title	Hire_Date	Location	Performan
0	EMP0000001	Joshua Nguyen	IT	Software Engineer	2011-08-10	Isaacland, Denmark	
1	EMP0000002	Julie Williams	Marketing	SEO Specialist	2018-03-02	Anthonside, Costa Rica	
2	EMP0000003	Alyssa Martinez	HR	HR Manager	2023-03-20	Port Christinaport, Saudi Arabia	
3	EMP0000004	Nicholas Valdez	IT	Software Engineer	2023-10-12	Port Shelbychester, Antigua and Barbuda	
4	EMP0000005	Joel Hendricks	Operations	Logistics Coordinator	2024-12-09	Lake Kimberly, Palestinian Territory	

In [109...

```
df.Country.value_counts()
```

Out[109...

```
Country
Congo                16286
Korea                16285
Sri Lanka            8409
Switzerland          8391
British Virgin Islands 8373
...
Indonesia            7983
Kazakhstan           7973
Montenegro           7972
Bhutan               7971
Palestinian Territory 7895
Name: count, Length: 243, dtype: int64
```

9. Is there any Correlation between Rating and Salary?

In [112...

```
cor = df['Performance_Rating'].corr(df['Salary_INR'])
cor
```

Out[112...

```
np.float64(-0.00020919799940916518)
```

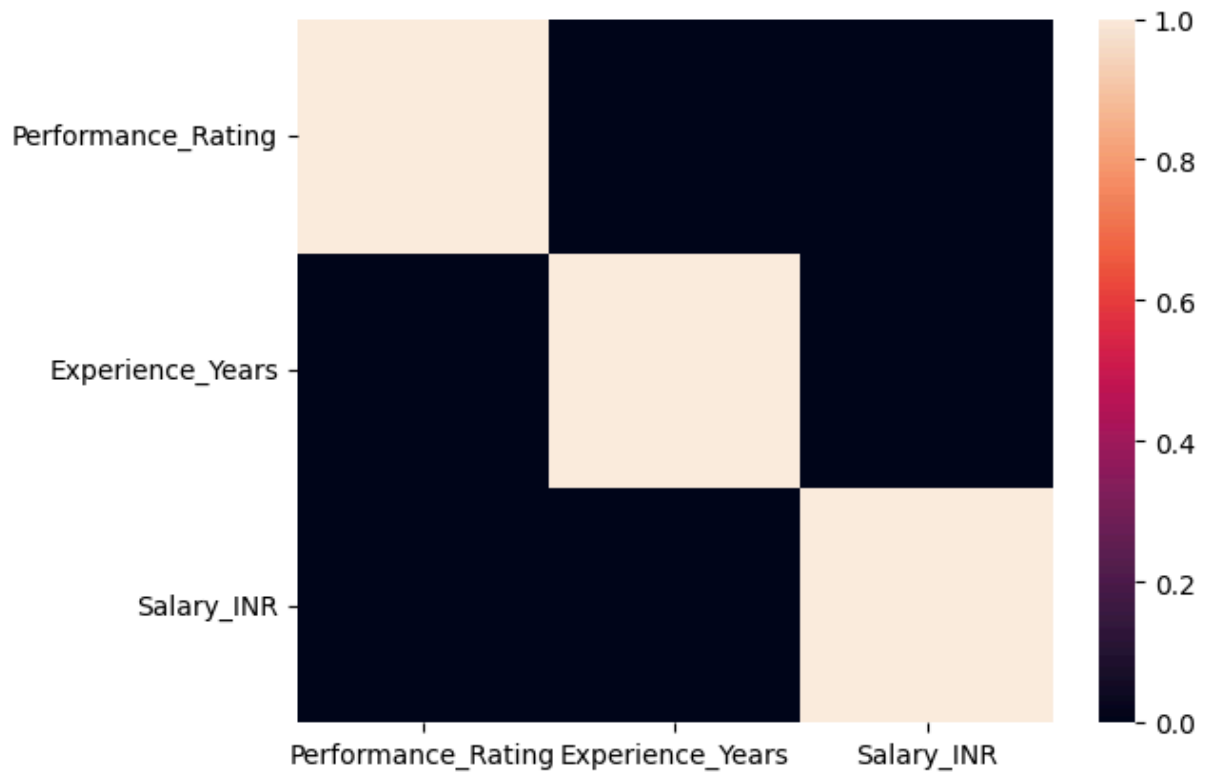
In [116...

```
df[['Performance_Rating', 'Salary_INR']].corr()
```

Out[116...

	Performance_Rating	Salary_INR
Performance_Rating	1.000000	-0.000209
Salary_INR	-0.000209	1.000000

```
In [118... sns.heatmap(df.corr(numeric_only = True));
```



10. How numbers of Hires has changed over Year/Time ?

```
In [123... df.insert(5, 'Hire_Year', df['Hire_Date'].dt.year)
```

```
In [127... df.drop(columns = 'Year', inplace = True)
```

```
In [128... df.head()
```

Out[128...

	Employee_ID	Full_Name	Department	Job_Title	Hire_Date	Hire_Year	Location
0	EMP0000001	Joshua Nguyen	IT	Software Engineer	2011-08-10	2011	Isaacland, Denmark
1	EMP0000002	Julie Williams	Marketing	SEO Specialist	2018-03-02	2018	Anthonyside, Costa Rica
2	EMP0000003	Alyssa Martinez	HR	HR Manager	2023-03-20	2023	Port Christinaport, Saudi Arabia
3	EMP0000004	Nicholas Valdez	IT	Software Engineer	2023-10-12	2023	Port Shelbychester, Antigua and Barbuda
4	EMP0000005	Joel Hendricks	Operations	Logistics Coordinator	2024-12-09	2024	Lake Kimberly, Palestinian Territory

In [132...

```
hire = df.groupby('Hire_Year')['Employee_ID'].count()
hire
```

Out[132...

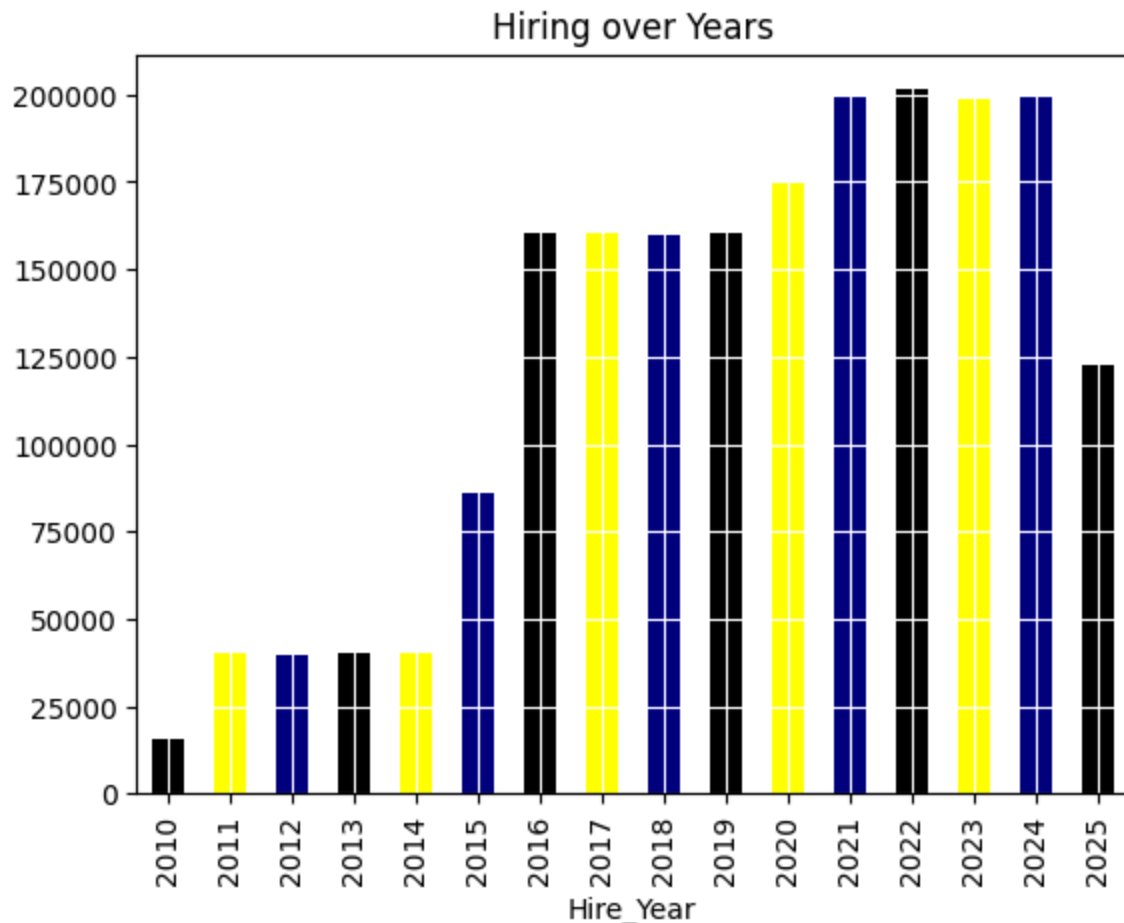
Hire_Year

2010	15520
2011	40089
2012	39765
2013	39988
2014	40202
2015	85984
2016	160249
2017	160363
2018	159658
2019	160202
2020	175460
2021	199366
2022	201373
2023	198982
2024	200001
2025	122798

Name: Employee_ID, dtype: int64

In [136...

```
hire.plot(x = hire.index, y = hire.values, kind = 'bar', color = ['black','yellow'],
plt.title('Hiring over Years')
plt.grid(color= 'w')
plt.show();
```



11. Compare the salaries of Remote VS Onsite employees

In [137... `df.groupby('Work_Mode')['Salary_INR'].mean()`

Out[137...
 Work_Mode
 On-site 896835.945792
 Remote 896965.326373
 Name: Salary_INR, dtype: float64

12. Show top 10 Employees from each Department with highest Salary

In [140... `df.groupby('Department').apply(lambda x : x.nlargest(10, 'Salary_INR'))`

C:\Users\hmmuz\AppData\Local\Temp\ipykernel_16380\105619432.py:1: FutureWarning: DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.
`df.groupby('Department').apply(lambda x : x.nlargest(10, 'Salary_INR'))`

Out[140...

		Employee_ID	Full_Name	Department	Job_Title	Hire_Date	Hir
Department							
Finance	888712	EMP0888713	Christopher Sloan	Finance	Finance Manager	2011-07-19	
	695808	EMP0695809	Todd Rodgers	Finance	Finance Manager	2019-12-27	
	459273	EMP0459274	Angela Payne	Finance	Finance Manager	2021-08-12	
	750893	EMP0750894	Nina Lara	Finance	Finance Manager	2021-10-19	
	780290	EMP0780291	Brittany Thompson	Finance	Finance Manager	2021-07-23	
...	...	...	...	...	...	...	...
Sales	86194	EMP0086195	Gabrielle Phelps	Sales	Business Development Manager	2015-11-23	
	1116580	EMP1116581	Kimberly Mullen	Sales	Business Development Manager	2025-01-09	
	1760918	EMP1760919	Christopher Farmer	Sales	Business Development Manager	2013-01-12	
	1878661	EMP1878662	Margaret Gardner	Sales	Business Development Manager	2025-04-23	
	1333220	EMP1333221	Benjamin Jones	Sales	Business Development Manager	2017-11-14	

70 rows × 13 columns



13. Identify the Departments with highest Attrition rate

In [143...

```
dept_reg = df.groupby('Department')['Status'].agg(Total_Emp = 'count', Resigned = 1)
dept_reg
```

Out[143...

	Total_Emp	Resigned
Department		
Finance	199873	40238
HR	159119	31736
IT	601042	119852
Marketing	240081	47793
Operations	300095	59397
R&D	99759	19919
Sales	400031	79725

In [146...

```
dept_reg['Attrition_Rate_%age'] = dept_reg['Resigned']/dept_reg['Total_Emp']*100
```

In [147...

```
dept_reg
```

Out[147...

	Total_Emp	Resigned	Attrition_Rate_%age
Department			
Finance	199873	40238	20.131784
HR	159119	31736	19.944821
IT	601042	119852	19.940703
Marketing	240081	47793	19.907031
Operations	300095	59397	19.792732
R&D	99759	19919	19.967121
Sales	400031	79725	19.929705